# Petition to list the Venus flytrap (Dionaea muscipula Ellis) as Endangered under the 1973 Endangered Species Act

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## Before the Secretary of the Interior

# Petition to list the Venus flytrap (*Dionaea muscipula* Ellis) as Endangered under the 1973 Endangered Species Act

**October 21, 2016**

**Figure 1**. A healthy Venus Flytrap individual growing in typically sandy soil and open conditions. Photo courtesy of James Fowler.

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**<u>Submitted this 21st day of October, 2016</u>**

Pursuant to Section 4(b) of the Endangered Species Act (“ESA”), 16 U.S.C. § 1533(b); Section 553(e) of the Administrative Procedure Act, 5 U.S.C. § 553(e); and 50 C.F.R. § 424.14(a), the individuals named above hereby petition the Secretary of the Interior, through the United States Fish and Wildlife Service (USFWS), to immediately protect the Venus flytrap (*Dionaea muscipula* Ellis) as a recognized Endangered Species and to quickly designate critical habitat for this plant species. The USFWS has jurisdiction over this petition. This petition sets in motion a specific process, placing definite response requirements on the Service. Specifically, the Service must issue an initial finding as to whether the petition “presents substantial scientific or commercial information indicating that the petitioned action may be warranted.” 16 U.S.C. § 1533(b)(3)(A). USFWS must make this initial finding “[t]o the maximum extent practicable, within 90 days after receiving the petition.” *Id*.

On behalf of the petitioners,

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## 1. Executive Summary

The Venus Flytrap (*Dionaea muscipula* Ellis) is a rare and localized species in the family Droseraceae endemic to wet coastal plain and sandhills habitats in North and South Carolina. Its dramatic specialized aerial snap trap makes it the most widely recognized carnivorous plant in the world. Its trap is a unique evolutionary adaptation shared with only one other species in the world. It is also evolutionarily unique as it is the only species in its genus and alone has an aerial snap trap. Wild populations of the Venus Flytrap are scattered, restricted in geography and suitable habitats, and declining quickly in number and size in response to illegal poaching, development pressures, and the ensuing loss of habitats with appropriate conditions including fire regimes. The Venus Flytrap is nearing extinction in the wild. Only 42 populations exceed a census population size of 500 individuals, and only some of these are fully viable over the longer-term. Together, these populations support less than 2% of this species' original population size.

The Endangered Species Act (ESA) states that a species shall be determined to be endangered or threatened based on any of five factors (16 U.S.C. § 1533 (a)(1)). The Venus Flytrap is threatened by at least **four** of these factors, warranting federal protection. First, it is threatened with significant **curtailment of habitat** resulting from development, fire suppression, vegetation encroachment, altered hydrology from wetland drainage, and soon sea level rise from climate change. Second, it is threatened directly by overexploitation from **poaching** and **seed collection**. Third, existing **regulatory mechanisms** have failed to protect the species. No habitat supporting larger populations is dedicated specifically to protect flytrap populations. The Venus Flytrap is not formally classified or protected by any federal entity. The decision by the U.S. Fish and Wildlife Service not to list the Venus Flytrap as Threatened or Endangered in 1993 has led to continuing declines in its range and abundance since then. Despite the fact that the Venus Flytrap was listed as a species of Special Concern-Vulnerable in North Carolina in 2010 and that poaching of this species became a felony in 2014, it continues to suffer poaching and population declines. This species is also threatened by **other factors** including climate change, storm surges, rising sea levels, the loss of genetic variation, and inadequate habitat connectivity, threatening its ability to sustain viable metapopulation dynamics.

The Venus Flytrap's geographically restricted range; limited population sizes; isolation of many of its habitats; lack of adequate habitat management in many occupied and potentially suitable habitats; continued losses to these habitats via woody plant encroachment, development, climate change, and rising sea levels; and continued poaching of plants from wild populations all threaten the persistence and viability of this plant species. The magnitude of these threats, their immediacy, and the taxonomic distinctiveness of the Venus Flytrap as the only species in its genus all argue for making the Venus Flytrap a priority for listing as **Endangered** under the Endangered Species Act.

"It may be counter intuitive to believe that a plant so often sold in grocery stores could be in trouble but the Venus flytrap truly is."

**In Defense of Plants –** http://www.indefenseofplants.com/blog/2015/10/6/the-plight-of-the-venus-fly-trap

## 2. Introduction

The Venus Flytrap (*Dionaea muscipula* Ellis) is probably the most widely recognized carnivorous plant in the world (Luken 2005). This iconic plant is alone in its genus, like its closest relative, *Aldrovanda vesiculosa* L., also in a monotypic genus. These species are the only descendant from a unique evolutionary event – the invention of an active snap trap. The great improbability of this surprising trait has captivated the curiosity and attention of people for centuries. Charles Darwin devoted considerable attention to this and other carnivorous plants because he realized that showing how natural selection could produce such remarkable and intricate adaptations would strongly support his theory. So many are beguiled by this plant and its ability to suddenly snap shut to imprison and digest its prey that this American plant has become the most widely recognized plant in the world. This popularity has also led to great demand for Venus Flytraps. Commercial growers now propagate, distribute, and sell these plants. More plants exist in captivity now than exist in the wild (Evans et al. 2012). Most plants in private hands derive from legitimate sources via propagation. However, market demand for this species and the lack of any formal certification program has also led many to poach plants from the wild for sale as house or medicinal plants.

Despite its popularity, few know that Venus Flytraps are highly restricted geographically. They only occur in the Carolina coastal plain with most sites within 90 km of Wilmington, NC, from Beaufort County NC to Charleston County SC (Mellichamp 2015). Even fewer are aware of how dependent this species is on having suitable habitats of a very particular kind available to it, namely wet pine savannas or pocosins that burn regularly or areas with similar conditions. Many of its populations have been lost to habitat development or conversion, shifts in habitat conditions, and overexploitation by plant collectors. Others have been eliminated by shade and competition accompanying the loss of fire from many traditional habitats. This species is also highly sensitive to changes in water levels as many populations lie only 2-4 m above sea level. These forces, alone and collectively, threaten the continuing viability of remaining populations and thus survival of this species.

Although steps have been taken to improve protection of certain habitats and populations and to limit the poaching of plants from wild populations, these efforts have not sufficed to arrest and reverse the slide of this species toward oblivion. This species therefore needs and deserves protection under the U.S. Endangered Species Act (ESA). Without such a listing, efforts to protect, sustain, and restore critical habitat for this species will not be systematic, sustained, and comprehensive enough to ensure survival of the Venus Flytrap.

In this Petition, we lay out what is known about this species' biology, ecology, biogeography, and conservation in order to document the past, present, and future factors that threaten this species. This information addresses the five criteria that the U.S. Fish and Wildlife Service uses to decide whether to add species to the list of federally threatened and endangered species. In light of the fact that the Venus Flytrap meets at least four of these five criteria, we argue that federal listing and protection are immediately needed to stem the strong historical declines observed in this species and to address the threats it faces. Without this listing and the resources listing provides, the Venus Flytrap will not be protected effectively throughout its limited range on the diverse land-ownership it occupies. Because this species faces substantial and immediate threats and represents the unique member of its genus and only one of two species in the world equipped with the snap trap adaptation, the Venus Flytrap should also be given priority for listing as an endangered species under the Endangered Species Act.

Listing the Venus Flytrap as federally Endangered is necessary in order to provide full **protection of remaining habitats**, resources and incentives to **actively manage remaining habitats**, and opportunities and incentives to actively **restore additional habitats and populations** in locations that are suitable for this species but currently unoccupied. In particular, ESA listing would ensure analyses and planning to promote adequate habitat and suitable conditions to sustain and restore flytrap populations throughout its former range. It would also allow forward planning to identify potentially suitable habitats based on projections of how the range of the Venus Flytrap will likely shift under projected future changes in land use, water levels, and climate.

## 3. The natural history and ecology of *Dionaea*

### 3.1. Taxonomy and varieties

*Dionaea muscipula* is a carnivorous plant endemic to the U.S. *Dionaea* is a monotypic genus of plant nested within the Droseraceae, a small family of 180 carnivorous plants located in the order Caryophyllales (Weakley 2015). The Taxonomic Serial Number for *Dionaea muscipula* is 22008. There are only three genera in the Droseraceae: *Drosera*, *Dionaea* and *Aldrovanda*, with the latter two being monotypic in having only one species each. These two genera also share the derived trait of having snap-traps, representing a unique evolutionary event (Cameron et al. 2002). *Aldrovandra vesiculosa* (the waterwheel plant) shares snap-traps with *Dionaea*, but its traps are smaller and underwater, being an aquatic plant. Droseraceae are sister to a clade consisting almost exclusively of other carnivorous plants, including Drosophyllaceae (monotypic *Drosophyllum*, a fly-paper plant not unlike Drosera from the western Mediterranean), Nepenthaceae (*Nepenthes*, with ca. 135 species, the tropical pitcher plants from SE Asia to Madagascar), and Dioncophyllaceae (monotypic *Triphyophyllum* from West Africa, a “part-time” carnivorous plant that has sticky leaves while a juvenile rain-forest vine, plus non-carnivorous *Dioncophyllum*) (Cameron et al. 2002).

The common name of *Dionaea* refers to Venus, the Roman goddess of love. The genus name, *Dionaea* ("daughter of Dione"), refers to the Greek goddess Aphrodite, while the species name, *muscipula*, is Latin for "mousetrap", reflecting its unique snap-trap (Wikipedia 2016). In 2005, the Venus Flytrap was adopted as the official State Carnivorous Plant of North Carolina (http://ncpedia.org/symbols/carnivorousplant).

There are 28 registered cultivars of *Dionaea* (Schlauer 2016, Carnivorous Plant Database; https://www.flytrapcare.com/venus-fly-trap-cultivated-variety-list). Many of these genetic variants are propagated vegetatively (asexually) via offshoots or tissue culture for sale as houseplants. Many vendors now offer flytraps from a variety of sources, making it difficult to trace whether these are legitimate and sustainable or derived from poached plants.

### 3.2. Evolutionary history

Although plant carnivory exists in at least 3 orders of higher plants, within the order Caryophyllales, genetic and fossil evidence suggest that carnivory evolved from one common ancestor (Cameron et al. 2002), an origin currently dated at 76 million years ago (Zanne et al. 2013). (See tree at: http://www.onezoom.org/OneZoom/static/OZLegacy/vascularplants_tank2013nature.htm). The Venus Flytrap represents an independent evolutionary lineage with a unique adaptation – active snap traps – that make it the most famous and emblematic carnivorous plant in the world. DNA sequence data confirm that *Dionaea* and *Aldrovanda* are sister genera that diverged from each other roughly 51 million years ago (Zanne et al. 2013) and together are the closest relatives of the sundews (*Drosera*) in the family Droseraceae (Cameron et al. 2002). This confirms that snap traps only evolved once in the history of plants – from flypaper-traps like those in present-day *Drosera, Drosophyllum*, and *Triphyophyllum*. Given contemporary distributions, ancestors of *Dionaea* presumably migrated to the Americas but then diminished in range over time to only occupy the Carolina coastal plain and Sandhills. Both genera are now monotypic with just one surviving species. *Dionaea* is thus taxonomically distinct, justifying priority consideration for listing.

Among terrestrial plants, *Dionaea* is unique in having rapidly closing snap traps. This trait evolved via several successive steps that involved evolving sensitivity to the presence of prey, rapid nerve-like signal transmission, curling leaf movements to encircle and digest prey and eventually a hinge to close around and imprison trapped prey (Gibson & Waller 2009). These adaptations include tentacles (common in *Drosera*) becoming modified into trigger hairs and marginal 'teeth', sticky tentacles being lost, digestive glands becoming depressed, and leaf movements evolving to become rapid. The snap trap itself may be an adaptation to snare larger prey that might otherwise tear away from sticky leaves and encapsulate them in a protected chamber from which few nutrients are lost. Their ability to trap larger prey and digest them efficiently for nutrients provides an advantage in terms of growth and reproduction in their wet, open, and highly nutrient deficient sandy soil habitats.

Their special adaptations have made Venus Flytraps of great interest to scientists since the days of Charles Darwin. New and innovative papers continue to be published on this species on their evolutionary history and relationships, physiology, leaf mechanics, and the biochemical mechanisms that allow them to propagate and respond to bio-electric signals so rapidly.

### 3.3. Species Description

Several descriptions of *Dionaea muscipula* exist including this one in Godfrey and Wooten (1981):

> Perennial carnivorous herb with leaves closely set on a short stem, the petiolar bases dilated and closely overlapping so that the short stem appears somewhat bulbous. Leaf blades hinged lengthwise in the middle, each of the two halves nearly kidney-shaped, the margin of each equipped with conspicuous bristles which interlock when the leaf is closed; petiole usually markedly winged and expanded distally. Inflorescence scapose, 1-3 dm tall, umbeliform-cymose. Sepals 5. Petals 5, white, spatulate, about 12 mm long. Stamens 15 (10-20). Ovary superior, 1-locular, style 1, stigma with numerous elongate papillae. Capsule ovoid. Seeds numerous, black and shiny, obovoid, tiny.

and this one based on Wikipedia's description:

> The Venus flytrap is a small plant whose structure consists of a rosette of four to seven leaves arising from a short subterranean stem that is actually a bulb-like object. Each stem reaches a maximum size of 3-10 cm, depending on the time of year (Rice 2007); longer leaves with robust traps are usually formed after flowering. Flytraps that have more than 7 leaves are colonies formed by rosettes that have divided beneath the ground.
>
> The leaf blade is divided into two regions: a flat, heart-shaped photosynthesis-capable petiole, and a pair of terminal lobes hinged at the midrib, forming the trap which is the true leaf. The upper surface of these lobes contains red anthocyanin pigments and its edges secrete mucilage. The lobes exhibit rapid plant movements, snapping shut when stimulated by prey. The trapping mechanism is tripped when prey contacts one of the three hair-like trichomes that are found on the upper surface of each of the lobes. The mechanism is so highly specialized that it can distinguish between living prey and non-prey stimuli, such as falling raindrops; two trigger hairs must be touched in succession within 20 seconds of each other or one hair touched twice in rapid succession, whereupon the lobes of the trap will snap shut, typically in about one-tenth of a second. The edges of the lobes are fringed by stiff hair-like protrusions or cilia, which mesh together and prevent large prey from escaping. These protrusions, and the trigger hairs (also known as sensitive hairs) are likely homologous with the tentacles found in this plant's close relatives, the sundews. Scientists have concluded that the snap trap evolved from a fly-paper trap similar to that of *Drosera* (Cameron et al. 2002).

### 3.4. Soils and climate

*Soils* – The Venus Flytrap only occurs in nitrogen and phosphorus-poor environments, such as bogs, pocosins, and wet savannas. It is restricted to wet sandy and peaty soils – particular conditions that maintain both hydration and nutrient scarcity. Because the lower Coastal Plain of the Carolinas is flat, losing <1m in elevation every 10 km, it has an abundance of habitats with wet, poorly drained soils. These result in numerous expanses of wetlands interspersed with ancient beach dunes and other areas of sandy soil. The resulting soils of the Coastal Plain are often sandy or mucky perched 2+m above an impermeable layer. The parent material here is generally unconsolidated coastal plain sediment deposited in the last 80,000 years (Daniels et al.1999; Soil Survey Staff 2016). In addition, excellent Flytrap habitats exist in areas of slightly higher topography (10-20m) on the coastal plain in SE North Carolina on the Lake Wales Ridge (R. Peet, pers. commun.).

Typical soil series in the middle and lower Coastal Plain include Baymeade (Arenic Hapludults), Blaney (Arenic Hapludults), Foreston (Aquic Paleudults), Grifton (Typic Orchraqualfs), Johnston (Cumulic Humaquepts), Kureb (Spodic Quartzipsamments), Leon (Aeric Haplaquods), Murville (Typic Haplaquods), Onslow (Spodic Paleudults), Pactolus (Aquic Quartzipsamments), and Woodington (Typic Paleaquults). While typical soils in the upper Coastal Plain include Blaney (Arenic Hapludults), Gilead (Aquic Hapludults), Rains (Typic Paleaquults) and Woodington (Typic Paleaquults).

The soils that support *Dionaea* are typically damp to wet and low to very low in nutrients and particularly phosphorus. An unresolved question concerns the degree to which aerially deposited nitrogen compounds (or other nutrients) might currently threaten *Dionaea* by accelerating the growth of competing non-carnivorous plants, allowing them to overtop and suppress the growth of *Dionaea* (see section 6.5). This potential nutrient enrichment effect makes maintaining adequately frequent fires (which volatilize nitrogen) even more important (see section 3.5). Fertilization experiments, however, suggest that *Dionaea* responds much more to phosphorus than to nitrogen (R. Peet, pers. commun.)

*Climate* – The Coastal Plain of the Carolinas has a humid-subtropical climate. Over the 30-year period from 1971 to 2000, the average annual temperature varied from 17.7°C in the lower Coastal Plain to 16.2°C in the upper Coastal Plain. Average daily maximum temperature was 24.2°C in the lower Coastal Plain and 22.6°C in the upper Coastal Plain, while the average daily minimum temperature was 11.1°C and 9.9°C, respectively. The mean annual precipitation ranges from 1140–1650 mm per year, with less precipitation falling in the Sandhills and more in the lower Coastal Plain (State Climate Office of NC).

### 3.5. Habitat needs and ties to fire disturbances

The Venus Flytrap is a habitat specialist adapted to highly particular habitat conditions that continue to lose key characteristics in many areas and have diminished greatly in extent. As with most carnivorous plants, it grows in habitats that are sunny, moist, and nutrient-poor, where it is likely to have a growth advantage over non-carnivores (Givnish et al. 1984; Givnish 1989). The Venus Flytrap inhabits flat, damp to wet, and open to semi-open habitats in the coastal plains of southeast North Carolina and a small area in northeastern South Carolina, usually on sandy nutrient-poor substrates often of low pH (see Soils above). In the lower Coastal Plain of North Carolina, most plants occur in wet pine savannas or wet pine flatwoods, typically occurring in ecotonal areas midway along a moisture gradient that spans pocosin shrublands on the wetter end to mesic to xeric pine savannas on the drier end (Roberts and Oosting 1958; Thornhill et al. 2014). They are also occasionally found at the ecotone between Carolina bays and adjacent plant communities. In the middle and upper Coastal Plain they occur on the narrow ecotones around Carolina Bays, streamhead pocosins (linear, evergreen shrub bogs along small creeks and their headwaters - NatureServe 2015), and sandhill seeps. In South Carolina, small populations of the species occur primarily at the ecotone between Carolina bays and adjacent plant communities in Horry & Georgetown Counties (Luken 2005). Because these communities are maintained by regular fires, retaining them will require us to apply regular burning.

The habitat specificity of the Venus Flytrap also reflects its association with particular micro-sites and transitional environmental conditions (ecotones). The ecotones that flytraps usually occupy can be quite narrow. In areas near Boiling Spring Lakes, NC, the elevation can change dramatically (up to 2m) over short distances (<100 m) from xeric longleaf pine-turkey oak savannas to lakes surrounded by pocosin shrubs. Here, the Venus Flytrap occurs across a narrow 4 m band where the most suitable moisture regime exists. If water tables shift, e.g., in response to drought, fire, or increased rainfall, Venus Flytrap populations must follow. For example, flytraps often die at drier ends of soil moister gradients, restricting their distribution to pocosin ecotones. The pocosin itself is usually too thick with plants to support flytraps, but occasional flytraps can be found perched on rotting, sphagnum-covered stumps further into the shrubs where wetter conditions prevail. This illustrates the importance of micro-topography to this species and how much it depends on particular moisture conditions. In another area (now protected as a conservation preserve), 30-year-old tire tracks from off-road vehicles created microsites for flytrap habitat. Where conditions are slightly dryer, flytraps occur in the ruts where wetter conditions prevail than in adjacent soils. In contrast, in wetter conditions, flytraps instead occur on adjacent soils next to, but not in, the ruts. The existence of local vegetation continua with subtle yet crucial shifts in structure help ensure that local conditions suitable for the Venus Flytrap remain even as the locations of patches of suitable habitat shift with changing precipitation, temperature, fire, and succession.

Habitats that support *Dionaea* are often insular, with plants restricted to narrow ecotones that burn regularly. Kologiski (1977) quantified the microhabitat of the flytrap in his classification of the vegetation types of the Green Swamp, North Carolina. He found that the plant is restricted to community types that are both wet and burn frequently (Kologiski Appendix II). He notes that flytraps often occur at 0-1% cover, but in 2 stands (147 and 148), flytraps occurred at 1-10% cover suggesting ideal conditions. Although Flytraps occur in 11 of ~100 plant community types (as defined by the U.S. National Vegetation Classification, Carolina Vegetation Survey data - [http://cvs.bio.unc.edu/](http://cvs.bio.unc.edu/)), all these reflect these same moist, infertile, open, and fire-maintained conditions (R. Peet, pers. commun.).

Ecotone habitats suitable for *Dionaea* were once (pre 1850) common in wet longleaf pine savannas, allowing it to become very abundant in this ecosystem that occurred commonly in southeastern North Carolina and, to a much lesser extent, in northeastern South Carolina. This is reflected in several quotes. In 1783, Young states (Coker 1928, p. 221): "I found this in great abundance in North Carolina and in some parts of South Carolina …"

Coker (1928) himself wrote that "In Hanover and Brunswick Counties . . the plants are so widely distributed and so abundant that they remain by the millions, although many of their stations have been destroyed…" Even in the 1970s, Kologiski (1977, p. 79) wrote that the flytrap "is found throughout the savannas in great abundance." The loss of the flytrap's natural ecotonal habitats, however, has increased greatly since then. In order to preserve the matrix of wet Longleaf Pine Savannas where flytraps generally occur, these habitats deserve critical habitat designation.

Plant cover in *Dionaea* zones is generally sparse with some grasses, herbs, and sphagnum but also bare patches between plants (Roberts & Oosting 1958). This generally reflects the effects of regular burning. *Dionaea* populations appear to thrive under fire regimes where fires recur at least every 3-5 years (Schnell et al. 2000; Luken 2005; Evans et al. 2012). Longer fire intervals are associated with rapid population declines. Thus, for populations of *Dionaea* to thrive and persist, fires must recur regularly and persistently. Sites without a long history of frequent fires rarely support *Dionaea*. In contrast, local plants can persist a long time if the site burns frequently. In the Green Swamp, *Dionaea* declined greatly after the Nature Conservancy replaced annual burns with a 3-4 year interval between burns (Palmquist et al. 2014). Frost 2000 concludes that *Dionaea* is "only found on the most fire-exposed areas of the landscape."

The dependence of *Dionaea* on fire reflects two key facts. First, this plant is a poor competitor for space against more conventional plants in crowded circumstances as it is short and its leaves are arranged in a flat rosette. In microsites with abundant plant cover, taller species overtop and outcompete *Dionaea* by shading it and burying its leaves in leaf litter. Taller surrounding plants may also reduce the ability of flytraps to catch larger insect prey (Schulze et al. 2001) or to benefit photosynthetically from the nutrients obtained from such prey (Givnish et al. 1984). Fires remove most of these competitors and the leaf litter, promoting flytrap growth, flowering, and seed production. Fires also provide the bare mineral soil conditions that promote seedling germination and establishment. Second, recurrent fire and flooding disturbances regularly kill plants and may eliminate populations. This makes *Dionaea* reliant on seed production and dispersal to suitable safe sites to sustain its populations. Given that its required micro-habitat patches are scattered and often small, its seeds must disperse in sufficient numbers to regularly colonize and recolonize these patches. In other words, this species necessarily relies on meta-population dynamics and can only persist in areas that sustain many suitable habitat patches in close proximity with fires frequent enough to maintain these conditions – but not so prevalent that all such patches are burned simultaneously. A corollary is that the persistence of *Dionaea* sub-populations requires regular and adequate seed production and dispersal to these suitable, but evanescent, habitat patches.

In summary, lack of regular fires and competition from taller plants are the two and interrelated major factors acting to limit the numbers and persistence of *Dionaea* within its wet longleaf pine savanna habitats (though hydrology also plays a role, as described below). The persistence of Venus Flytrap populations thus depends critically on having adequate soil moisture, low nutrient conditions, sparse woody plant cover, and the regular fires that reduce taller plant cover, oxidize nitrogen, and expose mineral (sandy) soils. In the absence of these effects, competing vegetation grows quickly to shade and suppress flytrap individuals, first suppressing flowering, fruiting, and growth, then killing the plants. It is thus necessary to maintain the more open, high light, and low-nutrient conditions that this species requires. Without fires, shrubs quickly overtake the narrow ecotonal bands where Venus Flytrap thrives, eliminating its habitats.

### 3.6. Reproductive ecology

*Flowering and Pollination* – There is little published about the pollination of Venus Flytraps and no documentation on what insects regularly or successfully pollinate this plant (Randall & Kunz 2016). A search of 350+ articles listed in Google Scholar revealed none that mentioned which species pollinate this rare plant. Roberts and Oosting (1958) did report that the species is "entomophilous, apparently by various beetles, small flies and possibly spiders, all of which may be seen in the flowers." The flowers are about 10-20 cm above the leaves and snap traps so insects pollinating the flowers do not get trapped. Venus Flytrap also exhibits temporal separation of traps and flowers; it temporarily suspends production of new trap leaves during flowering (Roberts & Oosting 1958).

*Seed set* – We found little information about how environmental conditions, plant size, or previous reproductive activity affect patterns of seed set in *Dionaea* (but see Section 3.7). We also could not find information on the regularity of seed set. No studies we are aware of document how far seeds disperse or the rates at which they colonize sites and successfully germinate in nature. Such studies are not simple but are greatly needed if we are to gauge how far populations can be from each other and still allow reciprocal recolonization and population persistence. Given that the frequency of such colonization events will depend on the seed rain, it also behooves us to learn more about how variation in seed set may influence local colonization dynamics.

*Seedling establishment* – Venus Flytrap seeds are reported to germinate either immediately after dropping from the fruits with no apparent dormancy (W. Morris, Duke University, pers. commun.) or later during the cool season (R. Peet, pers. observ.). This means that good years for seed production cannot serve to sustain recruitment in later years – except by allowing more plants to establish. Local recruitment and population persistence thus depend critically on maintaining mature reproductive individuals. Gibson (2016) further noted situations where seedlings establish in patches of bare sand being invaded by sphagnum mosses (also see Luken 2005). Larger adult plants appear to grow in drier sand without much sphagnum moss. Their local distribution may result from high seedling mortality during periods of drought. Adult flytraps appear to survive such periods much better, reflecting their deeper roots or the water they can store in their rhizomes.

Further work is clearly needed to learn more about how patterns of flowering, pollination, seed set, and seedling and population establishment in the Venus Flytrap affect local and regional population dynamics.

### 3.7. Demography

It is surprising to realize that the Venus Flytrap, a plant known around the world that has been studied for two centuries, lacks any comprehensive demographic study of natural populations. No such studies have been published. Quantifying the fundamental demographic processes of survival, growth, seed production, and the recruitment of new plants is clearly necessary if we are to understand how populations persist in the wild, how particular habitat or landscape conditions threaten population persistence, and how best to manage those threats. We do know from propagation studies that growers can produce flowering plants in three years. Without prey, plants would need longer (e.g., five years)

Despite the lack of comprehensive studies of Flytrap demography, a few studies exist that describe key demographic processes in Venus Flytrap populations and how these appear to relate to threats such as fire suppression and climate change. For example, Roberts and Oosting (1958) reported the results of a field experiment in which they translocated plants into uncleared and cleared plots to which ash (a potential source of nutrients) had been added (or not). They reported that removing competitors and adding ash (both consequences of fires) increased the probability of flowering. They also report that seeds rapidly lose viability at room temperature in the laboratory, with less than 2% of seeds germinating after only 100 days of storage. Beyond this study, we know of no published estimates of rates of fruit production, growth, or multi-year survival within natural populations. Estimates of all these are needed to understand how local populations persist.

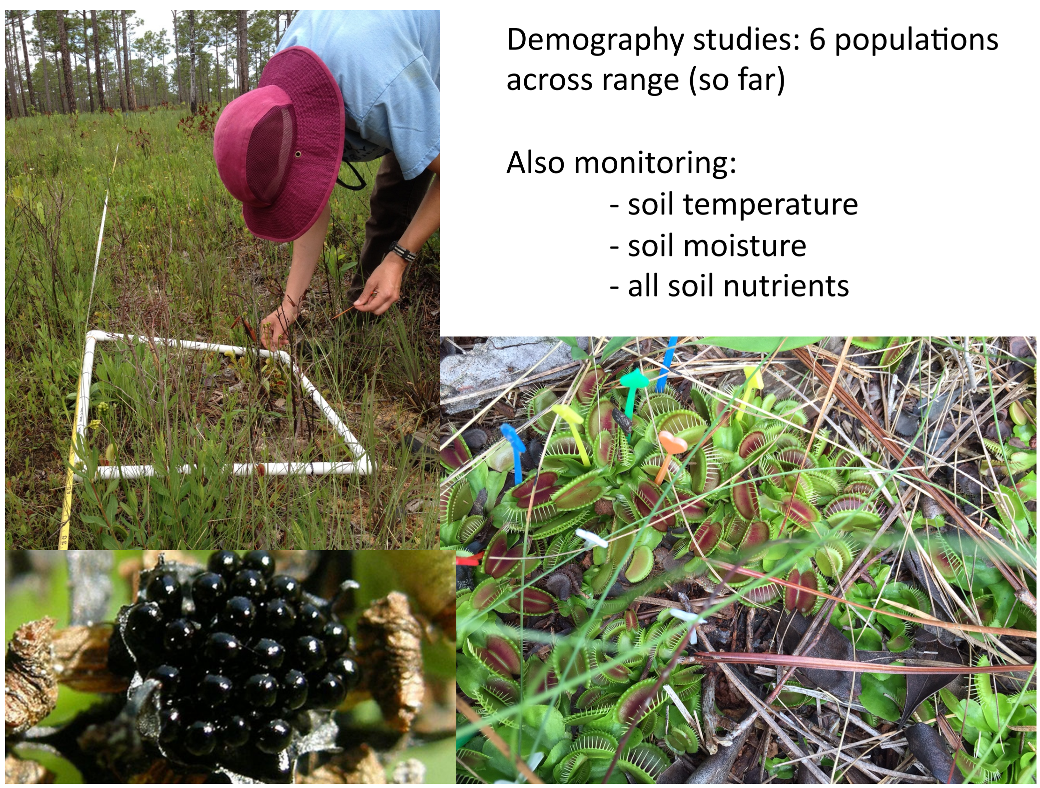


**Figure 2**. Photographs and description of the demographic monitoring of six Venus Flytrap populations initiated by Dr. W. Morris in 2015. Photos by Elsita Kiekebusch, North Carolina State University.

Fortunately, more comprehensive demographic studies are now underway. Starting in 2015, the laboratory of Dr. William Morris at Duke University (with funding from the

Strategic Environmental Research and Development Program at the Department of Defense) began a demographic study of six populations of Venus Flytrap in North Carolina (Fig. 2). Two of these are in the Croatan National Forest, two are at Camp Lejeune Marine Base, and two at the Ft. Bragg Army Base. The first annual re-census of these populations took place in 2016. All demographic rates (survival of individuals by size, growth in size from year to year, seed production, and recruitment of new individuals from seeds) are now being quantified for each population. This study is also measuring environmental variables that may affect these rates in every population. These include soil temperature, soil moisture, and soil nutrient levels. The populations being monitored were chosen to span a range of time since the last fire with two of the populations experiencing fires since the study began. This study will also complement and extend the clearing and ash-addition experiment of Roberts and Oosting (1958) by repeating their experiment at two sites and tracking the ensuing demographic responses (of all rates, not just flowering probability). Finally, these investigators are also planning experimental burns at Ft. Bragg, as well as greenhouse experiments to quantify the impacts of changes in soil moisture, soil nutrients, and prey availability on flytrap growth.

As these studies were initiated just over a year ago, no results have yet been published. Nevertheless, these studies complement this Petition and should be of great value to the U.S. Fish & Wildlife Service in providing key data to assess population growth and viability and the effects of management activities on local population dynamics and resilience. The data they generate can eventually be used to parameterize population models that will allow us to understand and predict how climate change and changes in fire frequency will affect flytrap populations into the future. These models will provide a useful summary of population dynamics. They will also help guide management, e.g., by allowing us to determine optimal fire frequencies. These results should be of great use to the USFWS for efficiently crafting an effective Recovery Plan once the species is listed under ESA.

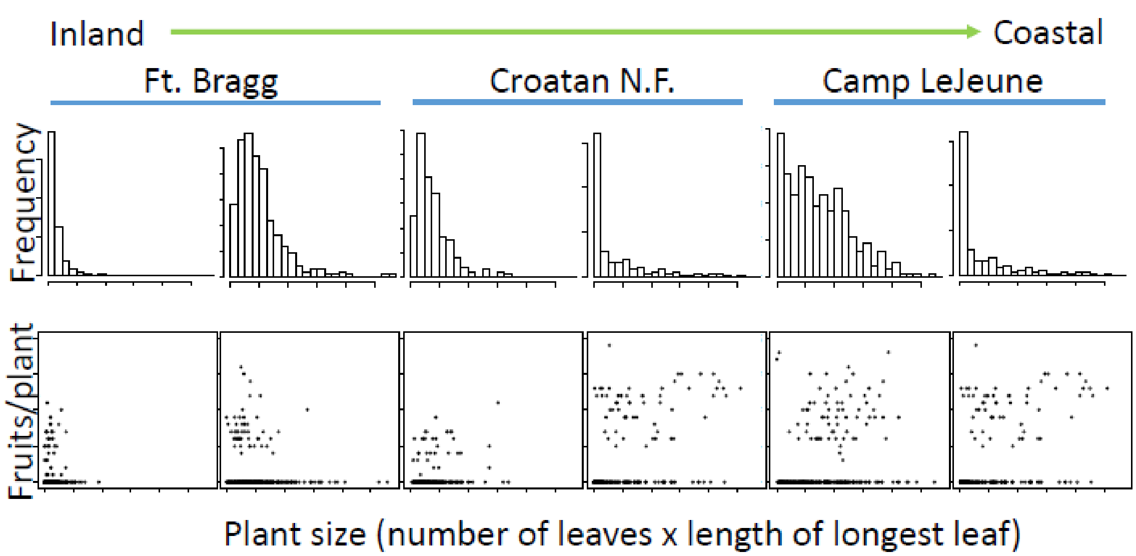

**Figure 3.** Distributions of plant sizes and the relationship between plant size and fruit production as observed in the six populations in North Carolina whose demographic behavior is being tracked (W. Morris, unpublished data).

Size distributions in many populations follow the classic J-shape expected in populations experiencing high attrition as seedlings mature to larger size classes (top row, Fig. 3). However, we also see individual populations at both Fort Bragg and Camp LeJeune that show substantially higher rates of apparent recruitment to larger plant sizes (assuming we can interpret this static pattern as reflecting a dynamic pattern through time). Over time, this study should allow us to determine the conditions that may be favoring this higher recruitment. In the lower row of Figure 3, we see the relationship between size and fruit production from the 2015 census of the six populations. Again, we see variation among populations in their size-dependent fruit production. This further suggests that differences among sites in environmental conditions are having demographic effects. While plants must reach a minimum size in order to fruit, fruit production appears not to increase with size once that size threshold for fruiting has been reached. Two of the size histograms show an under-representation of small plants (where the histogram is unimodal); these are long-unburned sites where competition is intense, fruit production is relatively low, and recruitment of new, small plants into the populations is correspondingly depressed.

At a broader geographic scale, we expect individual local Venus Flytrap populations to exist within the context of meta-population dynamics. As noted above, the microhabitat patches that Venus Flytrap plants require are scattered, sometimes small, and regularly affected by fluctuations in the water table and the time since fire (limiting the growth of taller plants). This requires that Flytrap seeds disperse in sufficient numbers to effectively colonize and recolonize newly available patches of suitable habitat. Thus, for populations of this species to persist in a given habitat, it must grow large enough to flower in a relatively short period of time, flower, produce seeds from those flowers, and disperse those seeds to other suitable microsites – all before local plants lose adequate water or are overtopped by competing plants. The Venus Flytrap is thus, per force, a species that depends on viable metapopulation dynamics. If populations are not regularly establishing at a rate sufficient to balance those being continuously lost, the species will not persist in that habitat. Both water and fire play key roles in maintaining the conditions that support this balance. Although it is usually difficult to estimate rates of population establishment and loss and to track which source populations donate seeds to colonize new habitats, the genetic tools we turn to next provide some hope that we may eventually be able to estimate these as well.

### 3.8. Population genetics

Given that small and dispersed populations now represent most of the total population of the Venus Flytrap with few large populations persisting in the wild, it behooves us to learn all we can about what population genetic variation exists in this species and how it is distributed within and among populations. Given that this species appears to rely on seed dispersal to colonize new sites, it may be subject to regular population bottlenecks that could restrict levels of population genetic variation. Likewise, the loss of many populations noted above and in Fig. 4 may reflect depletion of adaptive genetic variation and possibly locally adapted ecotypes. Population genetic variation can be very low in other rare species subject to similar meta-population dynamics tied to disturbance, e.g., Furbish's Lousewort (Waller et al. 1988).

As the Venus Flytrap is a species that is declining in the number of occupied sites and in population size within many sites, we should seek to learn all we can about the amount and structure of the genetic variation its remaining populations maintain. Population genetic surveys are needed to assess overall levels of genetic variability (affecting long-term adaptability and survival), any geographic variation that may be present (suggesting local ecotypes and/or the presence of unique genetic variants), and levels of inbreeding within individual populations. The amount and distribution of genetic variation affects a species' ability to adapt to local changes in environmental conditions. In addition, small populations are subject to genetic drift, the accumulation of deleterious mutations, fixation of the genetic load, and inbreeding depression reflecting the expression of deleterious mutations in homozygous combination (Keller and Waller 2002).

The North Carolina Botanical Garden and Carolina Center for Genome Sciences recently initiated a new project to sample *Dionaea* populations across its range for both tissue and seed germplasm (Randall & Kunz 2016). This broad survey will also apply a genomic approach ("Next Gen" RAD-seq) to evaluate DNA sequence variation within and among 50 Flytrap populations. The goals of this work include testing for isolation by distance, detecting sub-specific variation, and searching for evidence of reduced genetic variation and population bottlenecks, particularly in peripheral populations.

## 4. Geographic range and population status

### 4.1. The shrinking range of the Venus Flytrap

The Venus Flytrap has been found to occur naturally only in the Coastal Plain of southeastern North Carolina and far northeastern South Carolina (Fig. 4; Coker 1928; NC Natural Heritage Program Database 2014). Here, it is specifically native only to suitable habitats within a 130 km radius of Wilmington, North Carolina. These habitats used to cover thousands of square miles of the lower coastal plain of North Carolina (Kerr 1875). Most of the vegetation most critical to *Dionaea* has now been clear-cut, drained, and/or converted to agriculture or pine plantations (Kologiski 1977) resulting in the decline and disappearance of many populations (Figs. 4 & 5).

Herbarium specimens and historical records indicate *Dionaea muscipula* originally occurred at **259 sites** in 21 counties across North and South Carolina (Radford et al. 1968; NC Natural Heritage Program Database 2014). More recent observations suggest that the plant survives now as only **179 populations** in 12 counties (11 in NC and 1 in SC), with a notable collapse at the edges of the species' natural range (Figs. 4 & 5; NC Natural Heritage Program Database 2014; SC Heritage Trust Database 2016). Much of this collapse has occurred in inland populations close to development and population pressures. The loss of so many populations has had the consequence of increasing the isolation among remaining populations. Reductions in many local populations is likely to have concomitantly reduced the reproductive potential of these populations as remaining stands have become increasingly isolated from one another, limiting opportunities for pollination and increasing the likelihood for inbreeding and the loss of genetic variation (see sections 3.8 and 6.5). For example, the Sandhills populations, located on the Fort Bragg U.S. Army military installation, are now separated by more than 80 km from all other *Dionaea muscipula* sites (Fig. 4).

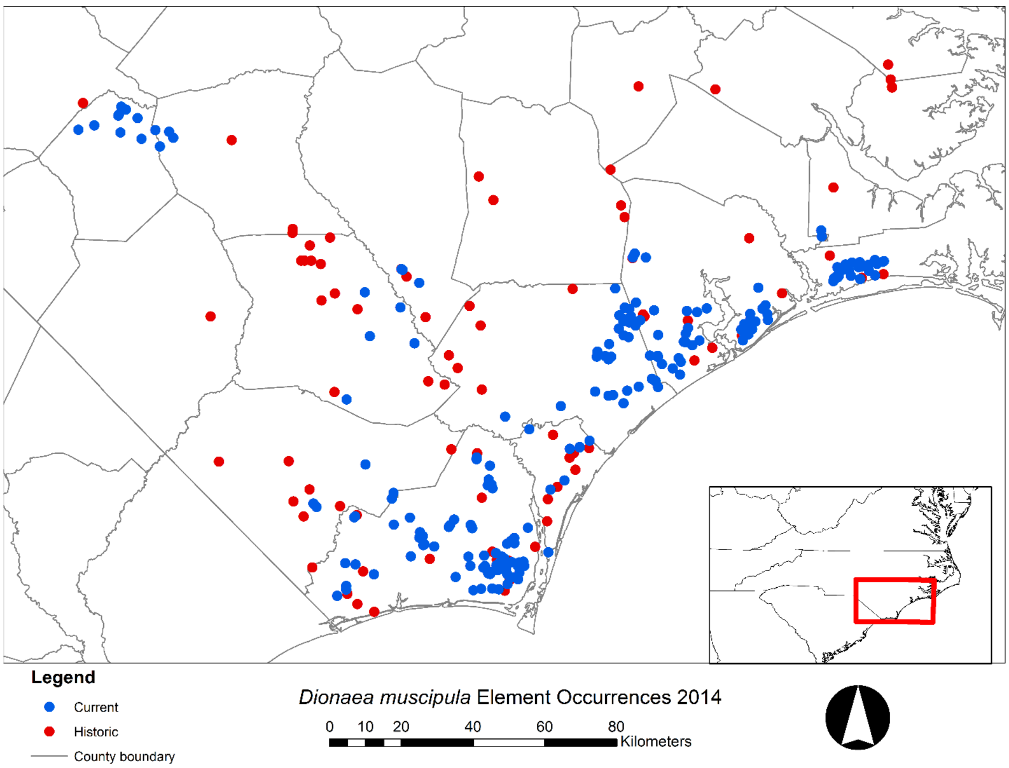


**Figure 4.** Current and historical range based on North Carolina Natural Heritage Program database from 2014 (*credit:* Yari Johnson). Data for South Carolina were unavailable despite several requests.

## 4.2. Population surveys and locations

Populations of the Venus Flytrap originally ranged from Pamlico County, North Carolina, south to Charleston County, South Carolina (an historic population only; Pittman pers. comm.). Among the remaining populations, 40% are very small containing only 10-100 individuals (North Carolina Natural Heritage Program, Fig. 5). Two exotic populations also exist, one in New Jersey and another in Florida (NatureServe 2015). These populations, however, are small and solitary. They are also useful, however, in that tracking any expansion in these populations may allow us to infer dispersal distances and perhaps other useful population parameters. Dr. Tom Smith at Florida State University is tracking the Florida population for just these reasons. Nevertheless, NatureServe (2015) concludes, "These extra-limital transplants are considered inconsequential to the conservation of the species."

Although most plants remaining in the wild are found on "protected" lands, only **16% of the total populations are considered to have minimum viable populations** (North Carolina Plant Conservation Program, unpublished data). This leaves the majority of populations at risk of extinction for stochastic reasons. Of the four main clusters of flytrap populations (Croatan National Forest, Marine Corps Base Camp Lejeune, Holly Shelter Game Land, and the Boiling Spring Lakes Preserve), only the Boiling Spring Lakes Preserve is managed for this plant species. Many of these populations occur in low-lying areas, making them vulnerable to storm surges and rises in sea level (see Section 6.5).

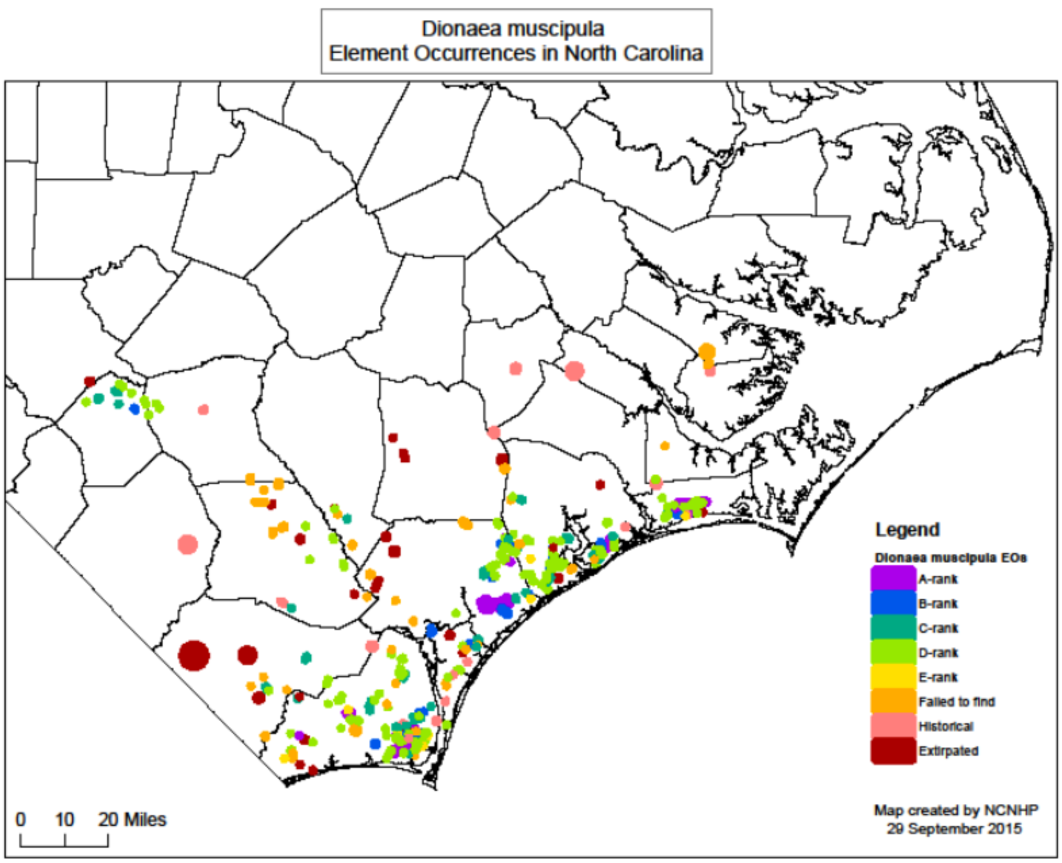


**Figure 5**. Current map showing the status of populations of *Dionaea muscipula*. The map shows extant, historical, and known extirpated populations across the geographic range of *Dionaea*. Note the loss of many populations (including its largest), the many small populations (ranks C, D, and E) and the scarcity and circumscribed range of healthy populations (ranks A and B). Source: Randall & Kunz 2016.

## 4.3. Record of population declines

Since the first range-wide inventory in 1958, **half** the known Venus Flytrap **populations have disappeared** (Randall & Kunz 2016), **including all those from 14+ counties**. Venus Flytrap populations continue to disappear at a non-sustainable rate, with a 23% decline in sub-populations occurring during a single recent ten-year interval that post-dates the US Fish and Wildlife Service's 1993 decision not to list this species (1992 – 2002, Randall & Kunz 2016). Populations continue to decline and disappear. In 2005, more than 100 flytrap population sites were identified in Southeastern North Carolina. By 2013, another 1/3 of extant populations had been lost, slipping to just 67 (North Carolina Natural Heritage Program). Of these 67 populations, only 42 populations exceed a minimally viable population census size of N = 500 individuals (Fig. 5; see also Fig. 88 in Evans et al. 2012). Of these 42, only nine (9) populations are judged to have "excellent viability" according to botanists with North Carolina's Natural Heritage Program. None of these populations are designated specifically for flytrap protection. Collectively these support less than 2% of its original population size (Evans et al. 2012).

It is also of great concern that populations continue to disappear even from public lands and in areas where botanists have carefully delineated populations and warned others about exactly where these populations are. In fact, there was even destruction of plants and whole populations on federal lands due to construction activities in 2016.

a) b)

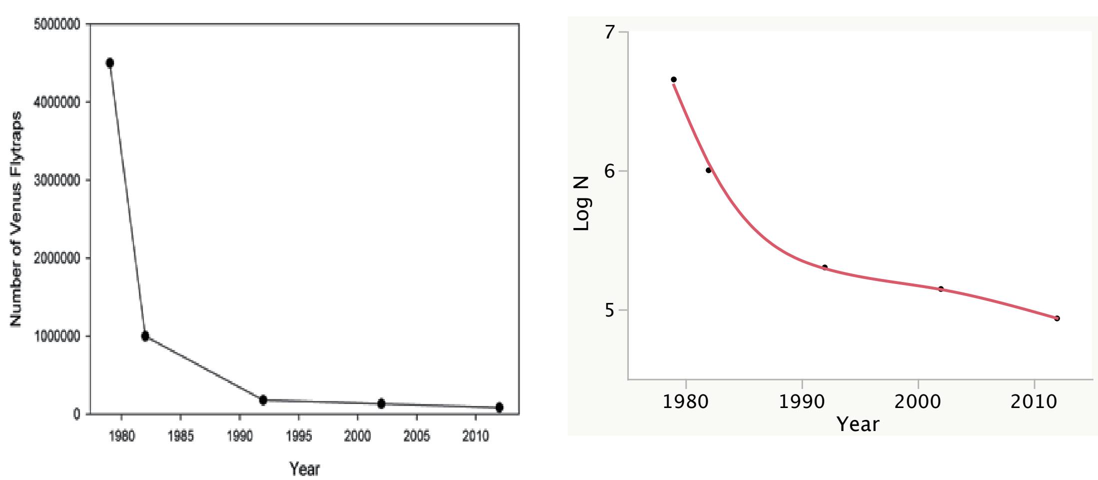


**Figure 6**. Declines in the total wild Venus Flytrap (*Dionaea muscipula*) population from 1979 to 2012 based on estimated sales (1979) and population counts (1982, 1992, 2002, 2012) from the North Carolina Plant Conservation Program. a) Linear scale, points connected by lines; b) Log scale, line showing smoothing spline fit (lamba = 100, $r^2 = 0.9$). Declines over the past 20 years have averaged 1.8% per year. Data used with permission from Evans et al. (2012).

As the number and extent of the remaining populations have declined (Fig. 5), so has the estimated total population size of the Venus Flytrap (Fig. 6). By 2015, according to Scientific American, fewer than 35,000 flytraps remained in the wild. As with declines in the number of extant populations, these declines in total population size are serious and sustained. Plotting these declines on both linear and log scales reveals both the overall dramatic decline in total population size (Fig. 6a) and the fact that high rates of total population decline continue through recent censuses (Fig. 6b). The rate at which this species' population is declining appeared to diminish in the late 1980s, just before the decision by the USFWS not to list this plant in 1993. However, the total population has continued to decline at an average rate of **1.8% per year** (average lambda population multiplier = 0.9818). This rate of decline corresponds to a half-life for the species of **38 years**, assuming that population and individual losses do not accelerate. Such a high rate of continuing decline in population numbers clearly signal that the persistence of this species is now threatened throughout its range.

## 5. Conservation status

### 5.1. Past conservation actions

Previously in 1993, the Venus Flytrap was considered for protection under the US Endangered Species Act. However, the US Fish and Wildlife Service determined that it was a “Potential candidate without sufficient data on vulnerability,” as published in Federal Register Volume 58, Number 188, Pages 51144 - 51190, September 30, 1993. Since then, considerably more data have been collected including the detailed data on population losses and population trends presented in Sections 3 and 4. These results strongly support this Petition. The additional demographic and genetic surveys now underway (see Sections 3.7 and 3.8) will fill additional gaps and would complement any U.S. Fish & Wildlife Service activities associated with listing this species, delineating critical habitats, and other steps to protect and manage its remaining habitats.

The U.S. National Wildlife Federation (2016) classifies the species as "**vulnerable**." Internationally, *Dionaea muscipula* was among 250 rare and threatened plants selected for inclusion in the first IUCN Plant Red Data Book published in 1978 (Kew 2016). *Dionaea muscipula* is rated as **Vulnerable** (VU) (A1acd, B1+2c) according to IUCN Red List criteria and considered **Vulnerable** (G3) under NatureServe criteria (Schnell et al. 2000, 2006). It was listed on **Appendix II of CITES** (Convention on International Trade in Endangered Species of Wild Fauna and Flora) at the 15th meeting of the Conference of the Parties (Doha, 2010). This group regulates international trade in threatened species.

Today Venus Flytraps only survive on sites that are owned and formally protected by The Nature Conservancy, the North Carolina government, or the U.S. military (Fig. 7). Nevertheless, losses continue from these sites (see p. 19 and below).

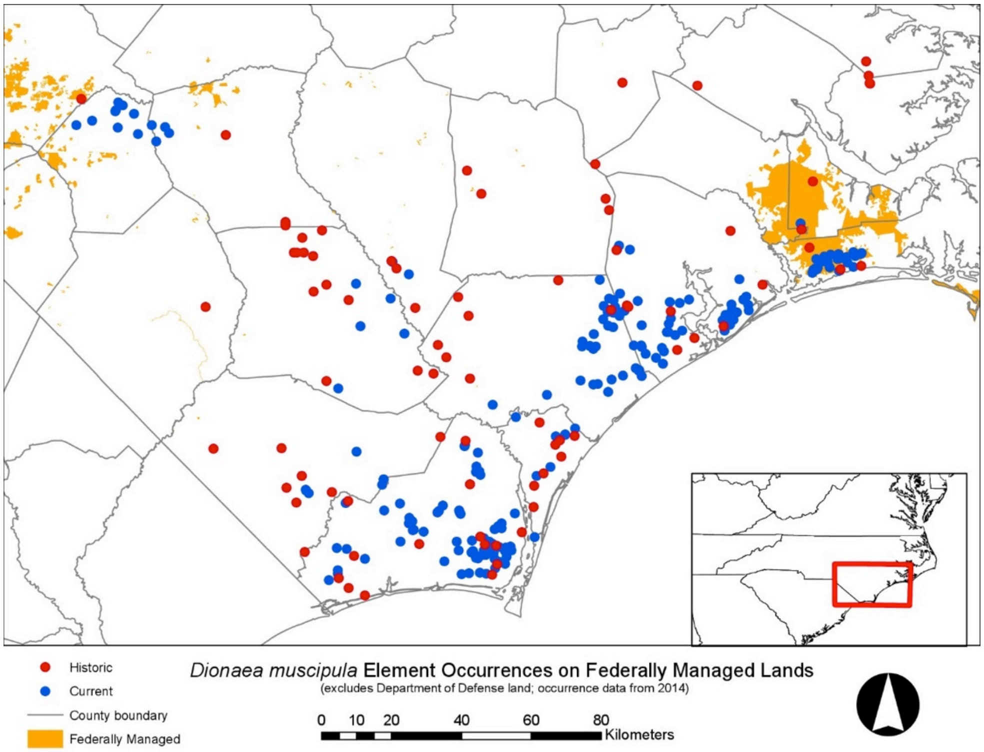


**Figure 7.** Current and historical range based on North Carolina Natural Heritage Program database from 2014 with non-Department of Defense federally managed lands. Most populations do not occur on federal lands and populations on federal lands continue to disappear. (*credit:* Yari Johnson)

### 5.2. Current regulatory context

Historically, poaching has been a significant threat to many populations, including some of those that were thriving the best and/or occurring on public lands (e.g., the Green Swamp). Many poachers, adept at avoiding detection, routinely robbed sites of 1000 plants or more (Platt 2015). Even when poachers were caught, penalties until recently were minor (e.g., a $50 fine) as the crime was only a misdemeanor. That is, collecting *Dionaea muscipula* from publicly or privately owned land without permission from the landowner was previously a misdemeanor in both North and South Carolina. Violators in South Carolina faced a maximum fine of $200 and up to 30 days in jail (SC ST SEC 16-11-590), while violators in North Carolina could receive a maximum penalty of sixty days in jail and a $1,000 fine. A civil penalty of $2000 could also be assessed for repeat offenders (NC GS § 106-202.19).

In 2013, a thief stole >1000 plants from a public park in Wilmington, NC. In 2014 another frequent thief was sentenced to 5.5 months in prison for stealing almost 300 plants from a National Park. Another man was sentenced to 80 days and a $1,000 fine (The Guardian, 27 July, 2016). Starting December 1, 2014, the penalty for poaching Venus Flytraps from public or private lands in Brunswick, New Hanover, Onslow, and Pender counties in NC was increased to become a Class H felony (NC GS § 14-129.3). This new law was exercised for the first (and to date only) time in 2015 when four men were arrested and charged with poaching 970 Flytraps from the Holly Shelter Game Land preserve in North Carolina (Platt 2015). This was roughly 3% of the entire wild population left of *Dionaea*. Two of these defendants received sentences of probation while 23-year-old Paul Simmons Jr. was found guilty and sentenced to 6-17 months in prison (with the fourth awaiting trial – The Guardian, 27 July, 2016).

As noted above, *Dionaea muscipula* is listed as a Species of Concern - Vulnerable by the North Carolina Plant Conservation Program and "Vulnerable" by the IUCN. Horticultural nurseries in North Carolina are thus required to obtain a permit before collecting propagation stock or offering propagated plants for sale. Permit applicants must meet the following criteria (Evans et al. 2012):

1. Identify the source of the initial stock of plants used for propagation;
2. Demonstrate that all protected plants to be offered have been nursery propagated and grown horticulturally;
3. Allow for a yearly inspection of site and facilities where protected plants are grown or stored for offer;
4. The person or persons offering protected plants shall maintain records of all acquisitions for the length of time these plants are in his possession. Such records shall be available for inspection by the Department and recorded on the permits or certificates of origin.

The US Fish and Wildlife Service lists Venus Flytrap as a **Federal Species of Concern** (FSC). This is an **informal term** and not defined in the federal Endangered Species Act. In North Carolina, the Asheville and Raleigh Field Offices of the US Fish and Wildlife Service define Federal Species of Concern as those species that appear to be in decline or otherwise in need of conservation and are under consideration for listing or for which there is insufficient information to support listing at this time. Subsumed under the term "FSC" are all species petitioned by outside parties and other selected focal species identified in Service strategic plans, State Wildlife Action Plans, or Natural Heritage Program Lists. On the state level, the Venus Flytrap is a candidate for state-listing through the North Carolina Plant Conservation Program (Randall & Kunz 2016).

***The above designations have conspicuously failed to ensure sufficient protection for the Venus Flytrap to sustain healthy and unmolested local populations.*** The persistence of the species is now threatened. To address these threats, the U.S. Fish & Wildlife Service should move immediately to list *Dionaea* as federally Endangered. Doing so would immediately expand protection for this species to the federal level under the 1973 Endangered Species Act, allow the designation and protection of critical habitat for this species, increase penalties and enforcement against poaching, and ensure more comprehensive and coordinated planning to protect and manage these populations and the habitats that support them.

## 5.3. Protecting the Venus Flytrap will also protect other rare and threatened species and habitat types

The Venus Flytrap occurs in particular habitats (Sections 3.4 & 3.5) that also support many other rare and/or threatened species. This means that protecting the habitats that support the Venus Flytrap would also automatically serve to protect many other species that depend on these habitat conditions and that may be at corresponding risk of sustained population declines and extinction. Quick action to strongly protect the Venus Flytrap and its habitats under the Endangered Species Act (ESA) could therefore serve to avoid endangering other species and having to separately consider listing them as Threatened or Endangered under the ESA.

No better icon symbolizes the beauty and exceptional floristic diversity associated with the Coastal Plain than the Venus Flytrap, "one of the most wonderful [plants] in the world" (Darwin 1875). This plant evolved unique tactics to survive in wet, nutrient poor soils in areas that experience frequent fires. Venus Flytrap habitat represents a unique hotspot for biodiversity along the East Coast based on the Rarity-Weighted Richness model for Critically Imperiled and Imperiled (G1 or G2) Species in the United States (NatureServe 2015). Annually burned wet pine savannas, which *Dionaea* favors and serves as a characteristic component, support the most floristically rich plant communities at small spatial scales documented in North America (Walker & Peet 1983; Peet et al. 2014).

The habitats occupied by Venus Flytraps support a broad diversity of other carnivorous plant species, native orchids, lilies, and other rare plants and several rare and/or endemic animals, including the Pine Barrens tree frog (*Hyla andersonii*). This diversity was recently formally recognized when the SE coastal plain was declared the newest biodiversity hotspot (#36) supporting >1800 endemic plant species (Noss 2016). This further highlights the need to preserve this habitat type and the many other plant and animal species that depend on it, many of which are in decline and vulnerable to extinction. Thus, protecting Venus Flytrap habitats could reduce the need to list these species as Threatened or Endangered in the future.

## 5.4. Critical habitat needs

The petitioners urge the Service to designate critical habitat for *Dionaea muscipula* concurrently with listing this species as federally Endangered. Critical habitat as defined by Section 3 of the ESA is: (i) the specific areas within the geographical area occupied by a species, at the time it is listed in accordance with the provisions of section 1533 of this title, on which are found those physical or biological features (I) essential to the conservation of the species and (II) which may require special management considerations or protection; and (ii) the specific areas outside the geographical area occupied by the species at the time it is listed in accordance with the provisions of section 1533 of this title, upon a determination by the Secretary that such areas are essential for the conservation of the species. 16 U.S.C. § 1532(5).

Congress recognized that the protection of habitat is essential to the recovery and/or survival of listed species, stating that: classifying a species as endangered or threatened is only the first step in insuring its survival. Of equal or more importance is the determination of the habitat necessary for that species' continued existence… If the protection of endangered and threatened species depends in large measure on the preservation of the species' habitat, then the ultimate effectiveness of the Endangered Species Act will depend on the designation of critical habitat. H. Rep. No. 94-887 at 3 (1976).

Critical habitat is an effective and important component of the ESA, without which the Venus Flytrap's chance for survival diminishes. The need to designate critical habitat for this rare plant is magnified by great declines in its range and number of populations (Figs. 4 and 5), its perilously low population size (Fig. 6), and the number and significance of the imminent threats it now faces (Section 6). Petitioners thus request that the Service propose critical habitat for this rare plant concurrently with its proposed listing. Some Venus Flytrap populations already exist on federally managed lands including the Croatan National Forest and military bases. Several Venus Flytrap populations occur on non-Department of Defense federal lands (Fig. 7).

### 5.5. Other urgent recovery actions needed

**Priority Category 1: Tasks needed to avoid imminent species extinction**

1. Protection of all remaining extant populations as an Endangered species under the ESA including full federal enforcement of statutes that preclude harm or damage to wild plants on public lands, poaching, and all activities that could threaten the extent or suitable conditions of habitats that support this species.
2. Directives to all federal agencies to protect, sustain, and actively manage *Dionaea* populations residing on federal lands. These directives should include provisions to monitor population sizes and habitat conditions to ensure continuing population viability and to guide adaptive management of these habitats.
3. Support to state and local levels of government to protect and monitor plants and populations on non-federal lands, promote public knowledge about these plants, and enforce all laws that serve to protect these plants and habitats.

**Priority Category 2: Tasks needed to maintain viable populations**

1. Support scientific surveys and studies of remaining wild populations in order to better understand its life history, demography, reproductive biology, population dynamics, and dependence on particular habitat conditions and disturbance regimes.
2. Use this information to design and implement active management activities on lands that support Venus Flytrap populations and, on an experimental basis, reintroductions or plantings onto other lands not currently occupied but judged to provide suitable habitat conditions.
3. Support scientific surveys and studies of the amount and distribution of genetic variation within and among populations. Use this information to assess the nature

and extent of genetic threats to individual populations and possibly the species as a whole and the corresponding advisability of efforts to "genetically rescue" populations deemed at risk of inbreeding or inviability and/or exchange genetic material more broadly among populations to ensure gene flow and adequate levels of genetic variability.

4. Support a public education campaign to alert citizens to the threats to this species and its habitats. Cooperate with and encourage botanical gardens and other public and private entities to promote conservation of this species by means of education, research, and efforts to actively conserve more habitats (e.g., via a surcharge on propagated horticultural plants to support *in situ* conservation).

## 6. Threats and criteria for listing

### 6.1. Losses of habitat and range

As noted above, the Venus Flytrap is a geographically restricted endemic species with a limited range (Fig. 4). Within this range, the Flytrap has suffered extensive levels of **habitat destruction and modification** that have eliminated many populations and greatly **curtailed its available habitat and range** (Figs. 4 & 5). These losses threaten its worldwide distribution and persistence, particularly in light of the additional threats enumerated above.

Its endemism makes the Venus Flytrap critically dependent on sustaining particular **wet pine savanna and pocosin habitat conditions** within its **limited range** on the Carolina coastal plains. Here, low-nutrient wet soil conditions and recurring fires serve to maintain the open, sun-lit, damp conditions that allow Flytraps to thrive. Sandhill Seep communities are found on wet sands underlain by clays on slopes in sand ridges or sandhill areas primarily in the Sandhills region, but are also present in scarps and sand ridges in the Coastal Plain (Schafale and Weakley 1990). Declines in Flytrap range and these suitable habitat conditions have contributed directly to the conspicuous population declines observed in this species. These dependencies make it critical to sustain hydrologic conditions and fire regimes similar to those that have sustained this species in the past.

Many wet, open habitats required by the Venus Flytrap have been encroached upon by shrubs and trees growing in response to **fire suppression**. Various authors (Frost 2000; Gray et al. 2003; Luken 2005) report that *Dionaea* populations thrive under frequent fire regimes and there are indications that with fire return intervals greater than 5 years, Venus Flytrap populations crash (NatureServe 2015). For this highly fire-dependent species, even fire return intervals greater than 2 years can result in population crashes after a period of approximately 5 years (Frost pers. Comm.). On Sandhills Seeps, Schafale (2012) notes:

> “These ecotonal communities are particularly subject to shrub invasion with inadequate fire, and are largely lost amid the expanding shrubs of adjacent Streamhead Pocosins in most places. It is somewhat unclear how extensive

> well-developed ecotonal seeps would be with a more natural fire frequency, but they would be much more abundant than they have become at present."

Community structure is strongly controlled by fire regime, and with fire these areas are open and herb dominated and somewhat resemble Pine Savannas but can quickly shift to shrub-dominated understory in the absence of fire (Schafale and Weakley 1990). Like other small natural communities in sandhill areas, nutrients mobilized by fire may be available to Sandhill Seeps even if they do not themselves burn (Schafale and Weakley 1990). Many of these Sandhill Seep areas are becoming overgrown with shrubs due to declining fire frequency:

> "Although collection and trade as a novelty item have contributed to the decline of *Dionaea*, its more fundamental problem is that faced by the great majority of Coastal Plain species in our area – destruction of habitat and fire suppression. In the fall-line Sandhills, *Dionaea* is now restricted to a very few sites on Fort Bragg; in the central Coastal Plain, it is also nearly extirpated. Substantial populations remain only in the Outer Coastal Plain, primarily in Brunswick, Pender, and Onslow counties." (Weakley 2015).

**Development** directly drives declines in Venus Flytrap populations by converting suitable habitats to parking lots, golf courses, and strip malls (Tucker 2010). Rates of development in the coastal plain of the Carolinas are high, increasing land values and the profits to be made from development. The recent loss of five additional Flytrap populations on federal lands occurring after reconnaissance and the explicit delineation of their locations (see Section 4.3) provides clear prima facie evidence that existing mechanisms to protect populations of this species are inadequate to prevent continuing erosion of populations to development.

Increases in roads, housing, and developed lands also threaten Venus Flytrap populations via a variety of indirect mechanisms. Most obviously, development acts to fragment and isolate remaining patches of natural habitat. Such regions then lose biological integrity and connectivity, limiting their ability to sustain Venus Flytrap populations. In particular, such regions become less able to sustain historically dominant fire regimes and correspondingly suitable conditions for the seed production (via successful pollination), dispersal (along corridors of suitable habitat), and colonization events that sustain metapopulation dynamics in this species. As populations become more isolated from each other, ecological "rescue effects" occur less often and gene flow is disrupted, increasing the likelihood of eroding genetic variation and inbreeding within these populations.

Another indirect effect of development is that it restricts our ability to sustain suitable habitat conditions for remaining flytrap populations. This occurs as natural hydrologic conditions are often disrupted as more impervious surfaces appear in the form of roads, parking lots, and buildings. Other lands may be drained for development. The construction of storm-water ditches and infrastructure further alters hydrology. Finally, land managers lose their freedom to conduct prescribed burns frequently and at suitable times as they seek to avoid generating in smoke in smoke-sensitive semi-urban areas. Habitats surrounded by more development are difficult to manage using fire (Evans et al. 2012). Thus even formally protected habitats become at risk of losing flytrap populations as development curtails management flexibility.

### 6.2. Over-utilization and exploitation

As already noted, carnivorous plants are of great interest to many collectors and aficionados. The Venus Flytrap is the most popular of all and is sold widely, mainly by commercial propagators and sellers. Some of these plants are propagated in captivity, usually asexually via tissue culture or more conventional vegetative propagation. Such propagation requires adequate facilities, skill, and time. This market is large, but most of the flytraps sold to amateurs last only a few weeks or months before dying because few have the resources, patience, and skill to keep these finicky plants alive. This sustains a steady demand for more flytraps, which maintains prices and thus pressure to poach plants from wild populations. More formal protection and regulation of what plants can be sold on the market could do much to reduce this over-exploitation of the dwindling number of wild plants and populations.

From the early 1990s to the present, North Carolina allowed a relatively constant number of commercial *Dionaea muscipula* sellers to collect plants. These nine documented nurseries do most of the business in propagating and selling Flytraps (Evans et al. 2012). Frantz (1991) estimated export totals of approximately **1,000,000 Venus Flytrap plants** per year. At that time, it is likely that many or most of these plants were of wild origin. The continuing demand for this species encourages poaching and even theft from nurseries. In May 2013, almost 90 percent of the Venus Flytraps in Wilmington's Alderman Park were stolen (~1000 plants). Also in 2013, 10,000 Venus Flytraps worth $65,000 were taken during a burglary at the Fly Trap Farm in Supply, NC. These and the recent theft associated with a felony conviction represent destruction of appreciable fractions of the total wild population of this species (Section 5.2).

Evans et al. (2012) note the strong negative impacts that poaching has caused. They state that it "…is still widespread and rampant and it is unclear if this problem can ever be completely removed." Poachers impose particularly strong mortality on the largest plants (Luken 2005, Evans pers. observ.). These large plants have many more flowers and fruits than smaller plants and thus contribute the most to the colonizations and metapopulation dynamics that sustain this species. When such size-selective harvesting is maintained over successive generations, we expect genetic changes to occur favoring flowering at smaller sizes instead of sustained vegetative growth. This could curtail demographic resilience as smaller plants with fewer flowers set fewer seeds. Such selective effects have been explored in populations of another exploited plant, American Ginseng (Mooney and McGraw 2007).

Because the Venus Flytrap can be readily propagated, the commercial propagation techniques and facilities that already exist for this species could be readily turned to the job of generating plants for out-planting to augment existing populations or establish new ones in patches of suitable but unoccupied habitat. However, without a formal Recovery Plan, there is little incentive or leadership to pursue this important work of actively re-introducing plants to restore existing populations and establish new populations. With formal listing and such programs, however, it should prove possible to quickly pursue active restoration and population supplementation in a way that could allow Flytrap populations to quickly rebound and recover. Thus, given what is known already about this plant and its habitat requirements, the U.S. Fish & Wildlife Service has the opportunity to move swiftly here to obtain a conservation success with this species.

### 6.3. Disease and predation

A thorough search of the literature reveals no information on the incidence of diseases on the Venus Flytrap, their consequences, or whether they represent any threat to population persistence. Similarly, we could find no information on the frequency, severity, or consequences of any herbivory on this species. This is to say simply that these potential impacts have not yet been examined or documented. The absence of evidence here should not be taken as evidence that such effects do not exist or are unimportant. Rather, further studies are needed to confirm their possible importance and how these threats may vary among populations and areas of its range.

### 6.4. Inadequate regulatory protection

Existing regulations and efforts to protect the Venus Flytrap have failed to halt sustained declines in populations of this species in recent decades (Figs. 2, 3, and 6; Evans et al. 2012). The population declines and losses documented in Sections 4.1 – 4.3 as well as recent poaching events and the 2016 loss of five more populations on federal land make clear that existing regulations and enforcement activities fail to address the full scope of forces threatening the persistence of this species. We have documented the several direct threats *Dionaea* faces as well as an even larger set of indirect threats. The number, complexity, and compounding and interacting nature of these threats demand an organized and comprehensive response of a kind that only the federal government can provide.

We also wish to stress, however, that the states of North and South Carolina have clearly demonstrated their commitment to protecting the Venus Flytrap and its habitats. Their many, substantial, and increasingly effective efforts at protection doubtless played an important role in slowing the rate of population decline (Fig. 6b). Working with other organizations like the North Carolina Botanical Garden, NatureServe, and state Natural Heritage Programs, they have also raised public awareness about the plant and the need to conserve its habitats and stem poaching. Increasing legal penalties for the poaching of flytraps and re-classifying this crime from a misdemeanor to a felony in North Carolina are likely deterring criminal trafficking in this species. More highly publicized cases could increase the effectiveness of this deterrent. Nevertheless, the evidence we have in our hands also plainly indicates that local efforts to stem poaching and protect populations have failed to address all the causes of population decline or stem these declines enough to prevent extinction. The number and scale of the efforts needed to protect, manage, and restore wild populations of the Flytrap simply appear to exceed the immediate capacities of local governments to adequately provide these.

Thus, **existing regulatory mechanisms** are **inadequate** to protect this species and the habitats it depends on. Only the federal government has the comprehensive mandate and resources to swiftly stem the decline of Venus Flytrap populations and thus avert extinction of this species. Insufficient resources and mechanisms exist at local levels to ensure systematic, inclusive, and sustained efforts to protect this species. It is thus incumbent on the federal government to meet these challenges and its obligations under the ESA by listing the Venus Flytrap as federally Endangered.

(Note: Listing under the ESA would activate the Lacey Act Amendment for plants. This would forbid the inter-state trade of endangered plant species. While this might appear to threaten the substantial horticultural trade in legitimately propagated Venus Flytraps, there are provisions of the Act that would allow this trade to continue in a regulated fashion. In particular, the horticultural industry could continue to sell Flytraps as a listed species with an interstate commerce permit. Commerce in several endangered plant species continues under this provision.)

### 6.5. Other threats to *Dionaea* survival

**Climate change** and **rising sea levels** represent additional immediate threats to remaining populations of the Venus Flytrap. That is, these factors could, alone and particularly in combination, further threaten to reduce or destroy populations or reduce the suitability of the Flytrap's remaining coastal plain savanna habitats, particularly in low-lying areas. Recent climate extremes are associated with recent increases in plant mortality and losses of habitat. Thus, it is susceptible to current and future climate change impacts. For example, the projected 0.5m rise in sea-levels in the 21$^{st}$ century represents a direct threat to Carolina Beach State Park and other remaining populations. As sea-levels continue to rise through the 21$^{st}$ century, low-lying coastal plain wet savanna habitats for the Venus Flytrap will be particularly vulnerable to flooding and to saline intrusion (Fig. 8).

We have recent very concrete evidence for the impacts that rising sea levels, increasingly severe storms, and the risks that storm surges pose for the Venus Flytrap in the form of Hurricane Matthew (see: https://www.washingtonpost.com/news/capital-weather-gang/wp/2016/10/11/horrific-rains-and-ocean-surge-hurricane-matthew-by-the-numbers/). In this storm, the rain that fell in the Southeast was equivalent to 13.6 trillion gallons of water, a volume equal to 75 percent of Chesapeake Bay (Fig. 9). The Lumber River in North Carolina reached a record 24 feet above its usual level, while the Tar River at Rocky Mount crested seven feet above flood stage. Storm surges broke records along the Carolina coasts with seven tide gages in the Southeast setting records including Hatteras and Wilmington, NC. The highest recorded storm surge was 7.8 feet above the ground in Fort Pulaski, Ga., near Savannah.

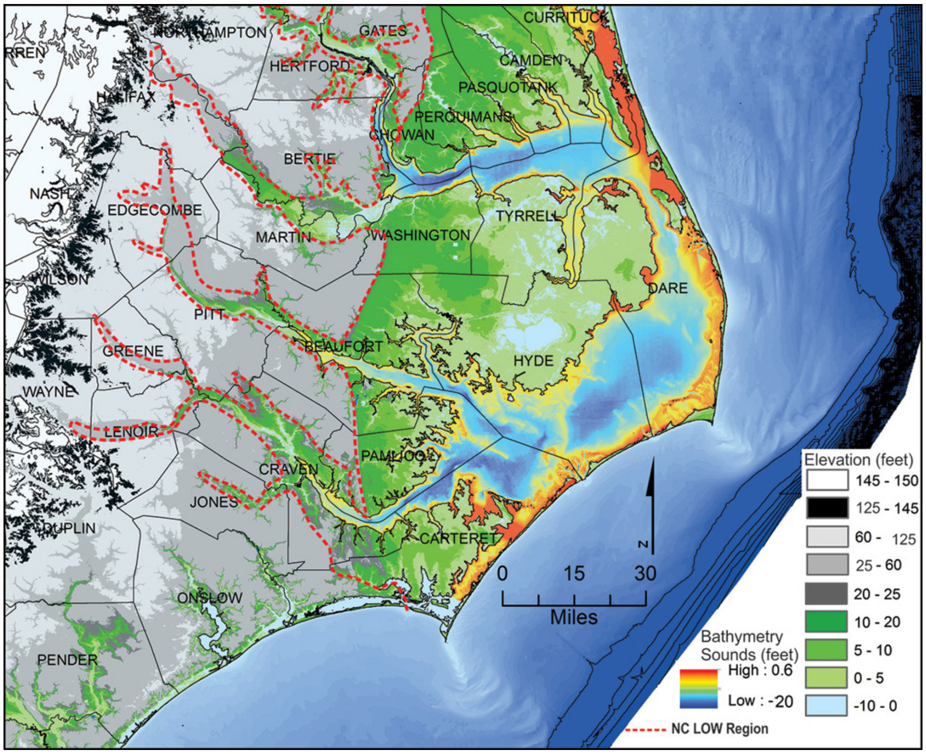


**Figure 8**. Elevation map of North Carolina's coastal plain where many *Dionaea* populations may be threatened by storm surges and rising sea levels.

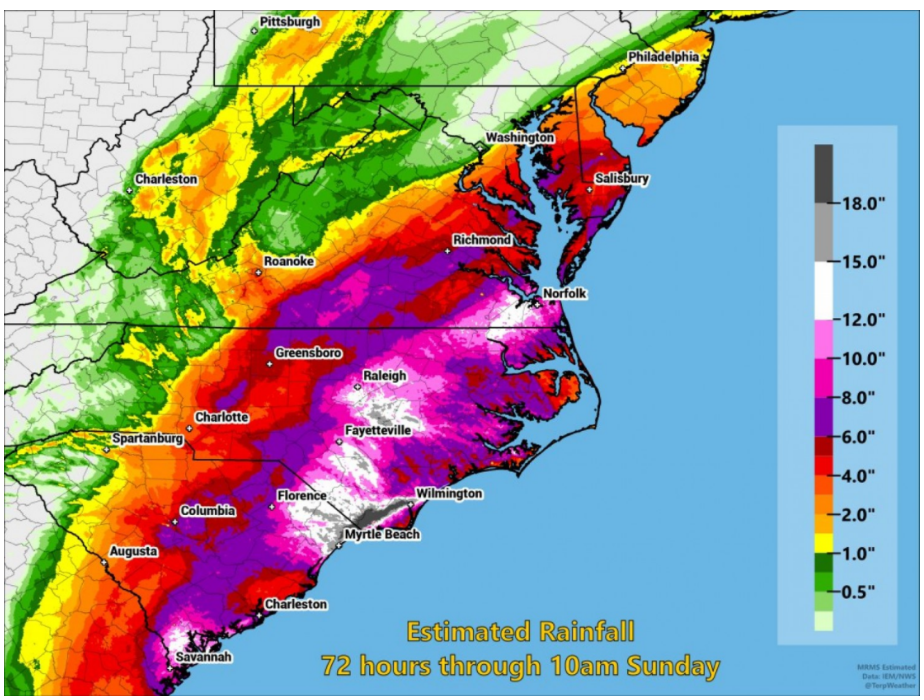


**Figure 9**. Rainfall associated with Hurricane Matthew, October 2016. In addition to these extraordinary rainfall levels, coastal areas were exposed to high storm surges that broke records in tidal regions along this entire section of coast. Venus Flytraps mostly occur <90 km from Wilmington, NC. Source:

https://www.washingtonpost.com/news/capital-weather-gang/wp/2016/10/11/horrific-rains-and-ocean-surge-hurricane-matthew-by-the-numbers/

It is too soon to evaluate the impacts these recent storms have had on Venus Flytrap populations, but they could be substantial if flooding associated with these record rains washed away individuals or even whole populations. Saltwater intrusion accompanying the storm surge may also have salinized some flytrap habitats, making them unsuitable for sustaining flytrap populations.

Atmospheric **nitrogen deposition** represents another potential threat to the persistence of Venus Flytrap populations. Nationally, an estimated 78 federally listed species are affected by nitrogen pollution (Hernández et al. 2016). Carnivorous plants are strongly adapted to growing in open conditions on soils that are highly nutrient deficient, e.g., the sandy soils that support Venus Flytrap populations (see section 3.4). Because they are short and light-demanding, Venus Flytrap plants are highly vulnerable to being overtopped by faster growing and/or denser populations of more competitive plants wherever higher even slightly higher nutrient levels support such growth. Plants adapted to growing at low nutrient conditions are thus most susceptible to nutrient additions as can occur via atmospheric nitrogen deposition (Stevens 2004; Giordani et al. 2014). These effects are associated with declines in nutrient-sensitive species and overall plant diversity in low nutrient habitats like chalk grasslands. Nitrogen deposition is already occurring at levels high enough to reduce plant diversity across much of North America (Simkin et al. 2016). Rare species appear to be particularly susceptible to nitrogen deposition. Suding et al. (2005) estimated the risk of species loss due to fertilization as >60% for the rarest species. Such "terrestrial eutrophication" likely poses an immediate threat to *Dionaea* populations in any areas where nitrogen deposition exceeds even 8 kg/ha/year from local (e.g., agricultural) and regional (e.g., industrial) sources. No one has yet tested for these effects, however. Threats to the Venus Flytrap would be compounded if ground- or surface waters are also becoming richer in nutrients. If such threats are affecting Flytrap habitats, it will be even more important to use controlled burns in these habitats to limit competitive growth and reduce nitrogen levels.

Finally, the Venus Flytrap may also be threatened by **genetic factors**. For example, if individual populations of the Venus Flytrap are highly adapted to very local conditions, changes in those conditions may result in declining levels of genetic adaptation that could impair individual and population viability. More generally, the ability of this species to adapt to changing conditions may depend on the levels of genetic variability it can sustain. For this species to survive, it must sustain genetic variation, adapt genetically, and be able to disperse to additional suitable habitats. Restricted population sizes, particularly in combination with limited levels of gene flow via pollen and/or seeds, will act to increase the rates at which genetic variation is lost via drift and individuals and populations inbreed. Inbreeding, in turns, acts overwhelmingly to reduce individual and population fitness as deleterious mutations are exposed in homozygous individuals – the phenomenon known as inbreeding depression.

Because we have no knowledge of levels of genetic variation, inbreeding, or inbreeding depression, we cannot provide any quantitative estimate of these risks. Nevertheless, we do know that small population sizes, uncertain reproduction, inadequate habitat management, and fragmentation and isolation of its habitats all act, individually and particularly in combination, to increase these risks by constraining local levels of genetic diversity and adaptability and by increasing levels of fixation within populations, homozygosity, and inbreeding depression. These genetic effects, in turn, likely impair individual fitness, population demographic resilience, and the ability of local populations to colonize new habitats. Thus, genetic factors pose additional potentially significant threats to this species.

## 7. Conclusion and summary

Despite being widely known, popular, and widely cultivated in captivity, wild populations of the Venus Flytrap are geographically highly restricted and have declined steeply in extent and numbers in response to illegal poaching, development pressures, and the loss of habitats experiencing suitable fire regimes. Given these declines and immediate threats, we petition the U.S. Fish and Wildlife Service to **list** this species as **Endangered** under the Endangered Species Act. To support this petition, we emphasize that:

- The Venus Flytrap is a highly localized plant with a specialized and taxonomically distinct evolutionary history.
- The Venus Flytrap has lost >95% of its original total abundance by losing many of its populations and having many others decline in size and viability. This has shrunk its range, isolated populations, leaving only 42 populations >500 individuals and 9 fully viable. Few of these most critical populations are adequately protected.
- Although the Venus Flytrap has been listed as "Vulnerable" (IUCN), a "Species of Concern", etc., these designations do not provide protection under the Endangered Species Act and have failed to protect this species from continuing declines in population number, extent, and viability since the USFWS considered this species for listing in 1993. Existing regulatory mechanisms have failed to protect this species and the habitats it depends on.
- The Flytrap's unique adaptation – aerial snap traps – have made it a widely celebrated carnivorous plant. This has added to the threats it faces by making it vulnerable to poaching and overexploitation. Without a program to certify seedlings as coming from legitimate sources and regulations on trading this species, economic pressures to poach wild populations continue. Listing the Venus Flytrap as federally Endangered or Threatened would enhance efforts to restrict this illegal poaching and trade in illegally collected Venus Flytraps.
- The Venus Flytrap is losing habitats and habitat conditions critical to sustaining its populations. Rapid economic development, shifts in hydric and fire regimes, and insufficient active management of its habitats all threaten to reduce or eliminate remaining populations. Listing the Venus Flytrap as federally Endangered would protect remaining extant populations from development and ensure suitable active management to sustain their viability. Actively managing remaining habitats and

working to protect and restore additional unoccupied habitats would protect both the flytrap and other rare species that depend on these habitat conditions.

- Climate change and rising sea levels further threaten Venus Flytrap populations by modifying or removing otherwise suitable coastal plain savanna habitats particularly in low-lying areas. Populations may also be vulnerable to genetic threats that have not yet been assessed.
- Listing the Venus Flytrap as Endangered would strengthen efforts by scientific and conservation organizations and many individuals to conserve Venus Flytrap populations.

Because of the perilous status of the plant's populations and the drastic decline in recent years, we ask that the Service protect the Venus Flytrap on an **emergency basis** while undertaking its status review.

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